Survival After Colorectal Cancer Surgery

Survival analysis

Study background

In this fictional study, a hospital research team is investigating survival after surgery for colorectal cancer. The study includes adults who underwent surgery with curative intent between 1 January 2019 and 31 December 2020. No patient had known residual cancer at the start of follow-up. Baseline information includes age, recorded sex, cancer stage, and comorbidity.

Follow-up begins on the surgery date. Vital-status information is available until death, the last date on which the patient was known to be alive, or administrative closure on 31 December 2022. The registry also contains information collected during follow-up.

Research questions

1. What is overall survival during the first two years after surgery, and how does it differ by cancer stage at surgery?

2. Which recorded patient information is associated with mortality during the first two years after surgery?

Choose and justify an analysis that addresses these questions. Present the relevant patient characteristics and findings, quantify uncertainty, and discuss the assumptions and limitations that matter for your conclusions. The choice of methods, representation of the data, model specification, and supporting checks is part of the assignment.

The dataset

The file `cancer.csv` will contain one row per patient. All dates use the format YYYY-MM-DD.

Variable	Description
`patient_id`	Patient identifier.
`surgery_date`	Date of surgery.
`age`	Age at surgery, in years.
`sex`	Sex registered as Female or Male.
`stage`	Cancer stage at surgery: I, II, or III.
`charlson_index`	Charlson Comorbidity Index assessed before surgery; higher scores indicate greater comorbidity.
`recurrence_date`	Date of the first documented recurrence during observed follow-up; missing if none was documented.
`date_of_death`	Recorded date of all-cause death; missing if no death was observed.
`followup_end_date`	Date of death for a patient with an observed death; otherwise the latest date the patient was known to be alive, no later than administrative closure.

Report and submission

Work in pairs. Present your analysis in the style of a small scientific paper (see template on the next page), with a brief methods paragraph, results, and a short concluding interpretation. Write a coherent account rather than separate answers to the research questions. Assessment focuses on the justification of your decisions, the main findings, and their interpretation, rather than the amount of output or whether results are statistically significant.

Submit a textual report as a PDF, not slides. The report must not exceed **1,200 words and four A4 pages**, with at most **three tables and figures combined**. A multipanel figure counts as one figure. Both length limits apply. The word count includes captions but excludes numerical table entries. The page limit includes everything. Do not submit an appendix or additional narrative material.

Submit a reproducible R or Python script separately, containing your analyses and checks, including those not selected for the report. The report is the primary assessed submission; the script may be checked to verify results.

Upload your report and script to Canvas before Monday 16 November, 09:00. Include the names of both students on the report.

Good luck!

General guideline

Write a coherent report rather than separate answers to the assignment instructions. Keep the research question and conclusion brief; dedicate most of the report to the methods and results. Please respect the limits with respect to the number of pages, tables, words, and figures. Replace the italic text with your findings.

1 Introduction

State the research questions in a few sentences. Identify what you are investigating and the population and outcomes of interest.

2 Methods

Describe the methods used to address the research questions. Explain the main analytical choices and their rationale, including how missing data were handled. Give enough information for the reader to understand the results.

3 Results

Begin with a concise description of the participants, relevant characteristics, and missing data.

Present the main findings for each research question in a logical order. Describe the direction, shape, magnitude, and uncertainty of the associations, supported by relevant numerical estimates.

Use tables and figures to support the narrative. Give them informative captions and refer to them in the text. Highlight what the reader should check in the figures and tables.

4 Discussion

Answer the research questions directly in a short paragraph. Explain what the findings mean and keep your claims consistent with what the study and analyses can establish. Do not introduce new results.